

Graphical Abstract: CAS-X Framework for Prioritizing Type 2 Diabetes Genes

CAS-X V5 PRIORITIZATION

Top Prioritized Genes

FTO | JAZF1 | VEGFA | NOTCH2 | TSPAN8

GWAS
50 T2D Loci

Gene Mapping
37 Genes

GTEx eQTL
Integration

Multi-Tissue
Support

Expression
Validation

External
Validation

Pancreas | Adipose | Muscle | Blood

GSE38642
GTEx Validation

GSE81608
scRNA Validation

Open Targets
8 Genes

CRISPR Support
5 Genes

Pathway Enrichment

Insulin Signaling | Type II Diabetes | PPAR Signaling

+175% Coverage Improvement Through Multi-Tissue Integration